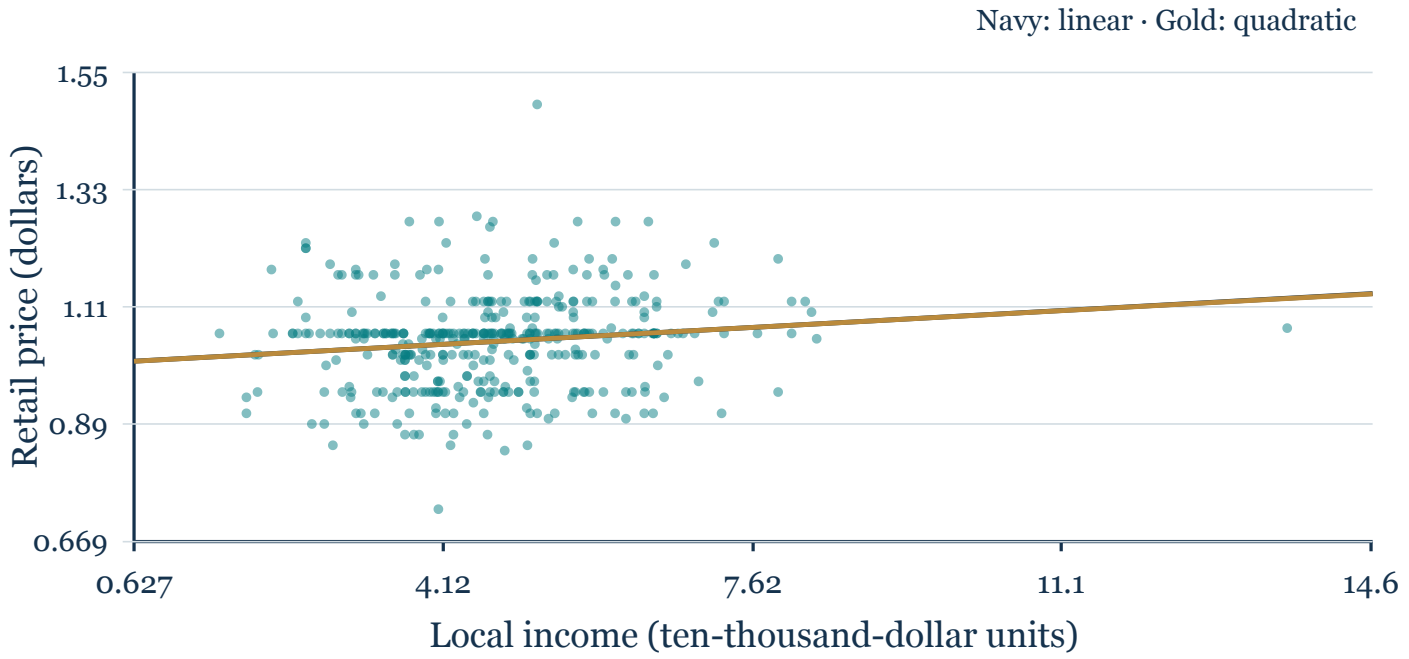
Do prices differ by the same amount or the same percent?

Choose a scale that answers the economic question.



401 observed positive-price and positive-income records. Lines use the actual level and quadratic regressions; no geographic sampling claim is made.

Define the observation and the usable sample

Inspect the data before interpreting a transformed variable.

> $X = \text{income}/10{,}000$

> Outcome: medium soda price, in dollars

A level–level model has a constant unit slope

A one-unit income change produces the same fitted dollar difference.

```
. regress sprice inc10, vce(robust)

------------------------------------------------------------------------------
             |               Robust
      sprice | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       inc10 |   .0090733   .0031523     2.88   0.004     .0028761    .0152705
       _cons |   1.002219   .0157154    63.77   0.000     .9713236    1.033114
------------------------------------------------------------------------------
```

$$\widehat{P} = a + bX$$

$$\Delta\widehat{P} = b\Delta X$$

A logarithm expresses a proportional scale

Small log differences approximate percentage differences.

> $\log(1) = 0, \quad \log(2) \approx 0.693, \quad \exp(0.693) \approx 2$

> $\log(2.20) - \log(2) = \log(1.10) \approx 0.09531$

> $100\{\exp(0.09531) - 1\} \approx 10\%$

A log–level model gives a proportional outcome contrast

Keep the regressor units visible.

> $\widehat{\log P} = a + bX$

> $100\{\exp(b\Delta X) - 1\}$

Which average does a log fit describe?

An average log is not the log of an arithmetic average.

> Arithmetic mean: $(1 + 9)/2 = 5$

> Exponentiated mean log: $\exp\{(\log 1 + \log 9)/2\} = 3$

> State which prediction quantity is being compared.

Exact log change

Try the comparison before opening the explanation.

Practice

In a log-outcome model, a policy dummy has coefficient 0.08. Give the usual approximate percentage contrast and the exact multiplicative contrast in the fitted exp(linear predictor).

Exact log change: explanation

Follow the calculation and interpret its units.

> Approximate: +8%. Exact: $100[\exp(0.08) - 1] = 8.3287\%$. This is a contrast in the exponentiated fitted log outcome. A conditional-mean claim in outcome levels additionally needs a justified retransformation treatment of the error distribution.

A level–log model gives a level outcome contrast

A percent change in X becomes a small change in log X.

$$\widehat{Y} = a + b \log X$$

$$\Delta\widehat{Y} = b \log(X_1/X_0)$$

A log–log slope is an elasticity

It relates proportional changes in both variables.

> $\widehat{\log Y} = a + b \log X$

> $\%\Delta \widehat{Y} = 100\{(X_1/X_0)^b - 1\}$

Elasticity

Try the comparison before opening the explanation.

Practice

In $\log(quantity) = a - 0.7\log(price) + u$, what is the local elasticity? What is the exact fitted quantity change for a 10% price rise?

Elasticity: explanation

Follow the calculation and interpret its units.

> The elasticity is −0.7. A 10% price rise multiplies fitted exponentiated log quantity by $1.1^{-0.7}$; the percentage contrast is $100[1.1^{-0.7} - 1]$, about −6.45%, not exactly −7%. Causal interpretation requires price exogeneity or a valid identification strategy.

Read the actual price–income elasticity

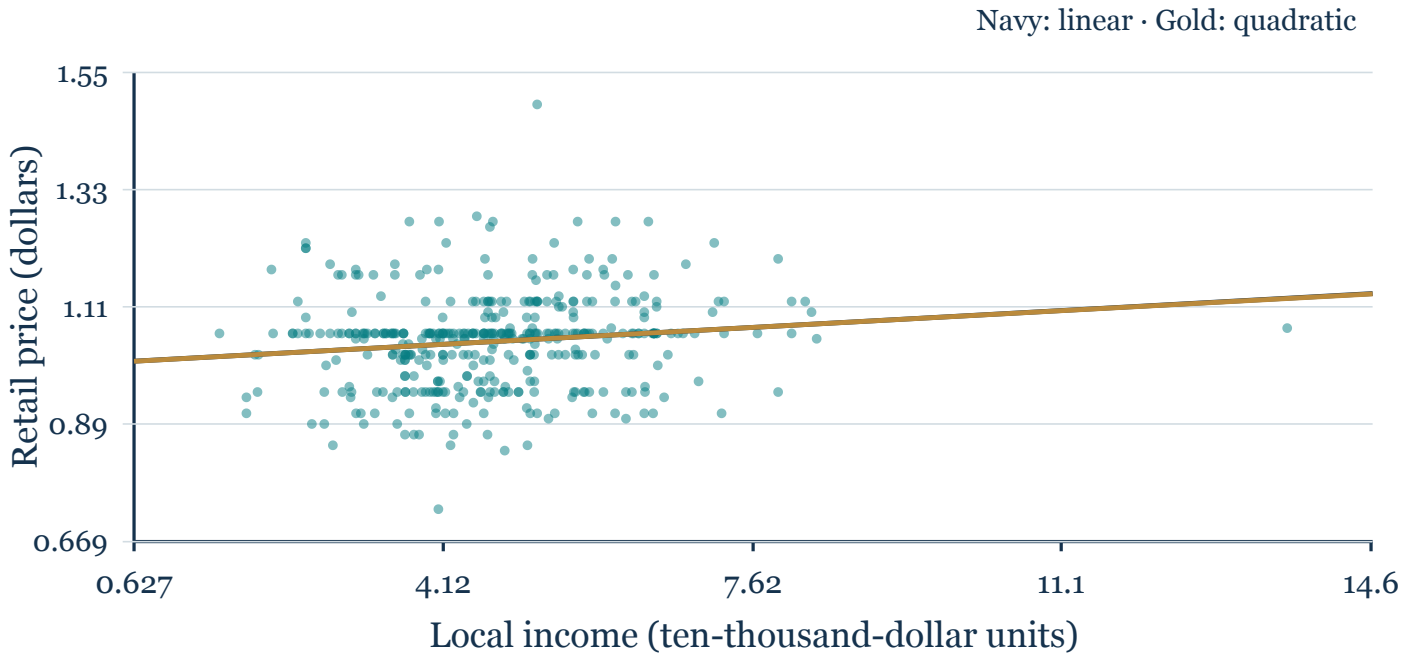
A unit-free slope still needs an empirical interpretation.

```
. regress lp li, vce(robust)

------------------------------------------------------------------------------
             |               Robust
          lp | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
          li |   .0374749   .0145083     2.58   0.010     .0089527    .0659971
       _cons |  -.3613775   .1559216    -2.32   0.021     -.667908   -.0548471
------------------------------------------------------------------------------
```

Compare predictions for the same realistic change

Use the equation, not just the coefficient label.



401 observed positive-price and positive-income records. Lines use the actual level and quadratic regressions; no geographic sampling claim is made.

A quadratic lets the slope change with X

Define the squared term before interpreting it.

> $\widehat{Y} = 500 + 8X - 0.2X^2$

> Local slope: $d\widehat{Y}/dX = 8 - 0.4X$

> General slope: $b_1 + 2b_2X$

Use a finite quadratic difference for a finite change

The tangent slope is a local description.

> $\widehat{Y}(10) = 560, \quad \widehat{Y}(15) = 575, \quad \Delta\widehat{Y} = 15$

> $\Delta\widehat{Y} = b_1\Delta + b_2(2X\Delta + \Delta^2)$

> Local approximation: $4 \times 5 = 20$, not the finite difference.

A turning point must be inside a meaningful range

Do not report an unsupported optimum.

> $X^* = -\dfrac{b_1}{2b_2}$ when $b_2 \neq 0$

Quadratic slope

Try the comparison before opening the explanation.

Practice

For $\widehat{score} = 500 + 8X - 0.2X^2$, give the slope at $X = 10$ and the turning point. What must be checked before using the turning point?

Quadratic slope: explanation

Follow the calculation and interpret its units.

> The slope is $8 - 0.4X$, which equals 4 at $X = 10$. The turning point is $X = 20$, a maximum because the squared coefficient is negative. Check that $X = 20$ lies within the observed and substantively meaningful range; extrapolated turning points are unreliable.

Inspect the actual curvature rather than assume it

A flexible model may provide little evidence of bending.

```
. regress sprice c.inc10##c.inc10, vce(robust)

------------------------------------------------------------------------------
             |               Robust
      sprice | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
       inc10 |   .0091573   .0103187     0.89   0.375    -.0111286    .0294432
             |
c.inc10#c.inc10 |  -7.86e-06   .0008332    -0.01   0.992    -.0016458    .0016301
             |
       _cons |   1.002011   .0304353    32.92   0.000     .9421772    1.061845
------------------------------------------------------------------------------
```

Ask Stata for slopes at stated values

margins makes the comparison explicit.

```
. margins, dydx(inc10) at(inc10=(2 4 6))

Conditional marginal effects                                    Number of obs = 401
Model VCE: Robust

Expression: Linear prediction, predict()
dy/dx wrt:  inc10
1._at: inc10 = 2
2._at: inc10 = 4
3._at: inc10 = 6

------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
inc10        |
         _at |
          1  |   .0091259   .0071961     1.27   0.205    -.0050213    .0232731
          2  |   .0090945    .004394     2.07   0.039      .000456    .0177329
          3  |    .009063   .0030074     3.01   0.003     .0031507    .0149754
------------------------------------------------------------------------------
```

Retransforming a log fit needs care

The exponential of an average log is not the average level.

> $\log Y = m(X) + u$

> $E(Y \mid X) = \exp\{m(X)\}\, E\{\exp(u) \mid X\}$

> The retransformation factor need not be one or constant across X.

Zeros and negative values are substantive information

Do not hide them through an automatic log transformation.

> The natural log is defined only for positive values.

> In this retail application, the analysis explicitly requires positive observed price and income.

Choose a form for a decision-sized comparison

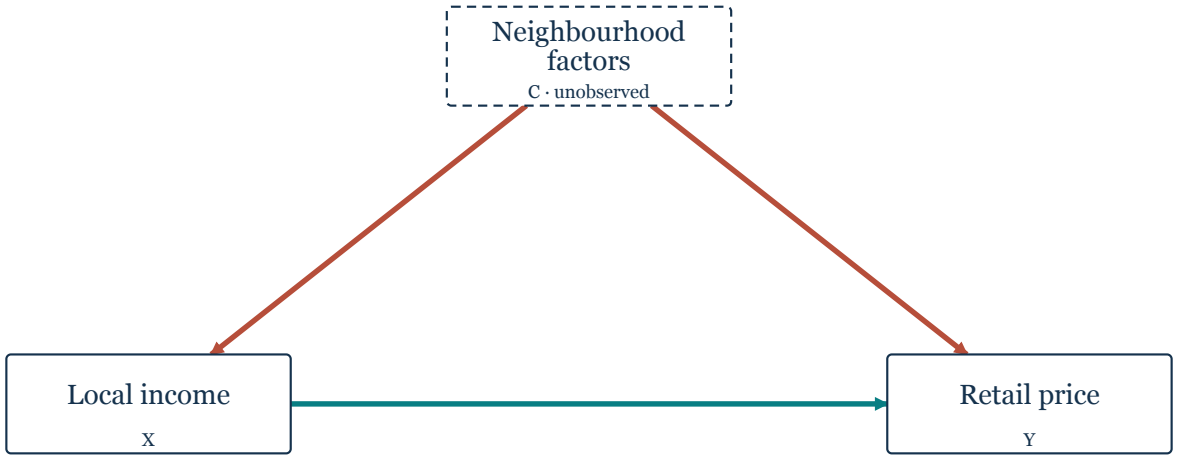
Keep fit, interpretation and design connected.

> An analyst studying affordability may want dollar price differences.

> Use graphs and stated prediction contrasts to communicate the model.

Transforming the variables does not close a backdoor

A better scale and a credible comparison answer different questions.



1 open directed route · 1 open alternative · gold: conditioned on

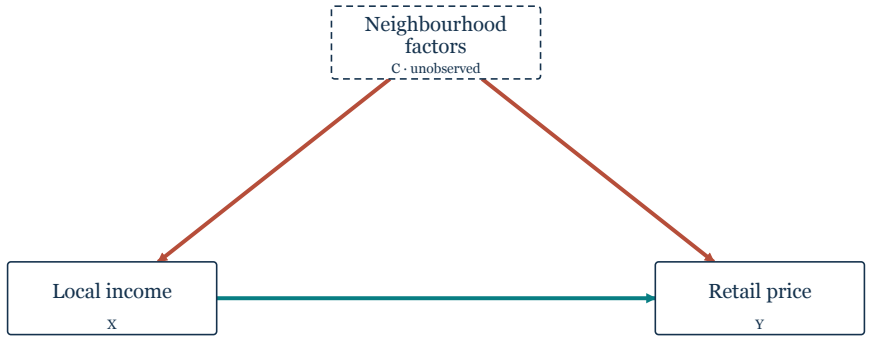
Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

> Functional form chooses the scale and shape.

> Identification addresses the alternative causal explanations.

DAG practice: logs and a common cause

Trace the routes before choosing a control.



1 open directed route · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Practice

Neighbourhood characteristics influence local income and retail prices. Would replacing prices and income by their logarithms close the common-cause route? Explain the distinct roles of the graph and the functional form.

DAG practice: logs and a common cause: explanation

Explain what conditioning changes in this stated graph.

> The route $X \leftarrow C \rightarrow Y$ remains. A numerical transformation does not remove the economic mechanism represented by C. A better fit or a convenient elasticity therefore does not establish causal income variation. The graph states the proposed causal relationships and helps assess controls or alternative designs. The equation chooses how a given comparison is represented numerically and which finite or local contrast is reported. An applied analysis needs both a defensible comparison and a useful specification.

Translate retail coefficients into a useful contrast

Reproduce the evidence, then explain its economic meaning.

Practice

Run meeting_05.do using retail.csv. In the level model, calculate the fitted soda-price contrast for a 20,000-dollar income difference. In the log–log model, calculate the exact contrast for a ten-percent income increase. State the units and the prediction quantity in each answer.

Translate retail coefficients into a useful contrast: explanation

A reproducible result needs both a calculation and an interpretation.

> Income is rescaled by 10,000 in the level model, so the proposed change is two units. The fitted price contrast is two times 0.009073, or 0.018147 dollars. Treating the coefficient as dollars per one dollar of income would give the wrong magnitude. The log–log income slope is 0.037475. A ten-percent income rise multiplies the exponentiated fitted log price by 1.1 raised to that slope, giving 0.3578 percent. This is a contrast in exponentiated fitted log values. A claim about arithmetic mean prices also requires the retransformation factor to be handled appropriately. These local-income associations do not identify a causal income effect.

Connect functional form with the next data-quality question

A better curve cannot repair every data problem.

> We can now translate coefficients in levels, logs and quadratic models into meaningful changes.

> The next lesson asks what happens when we collect more observations while the design or measurements remain imperfect.